APPENDIX - AI USE DOCUMENTATION

I. Deepseek

1. AI Tool Information

- Tool name: DeepSeek
- Role simulated: Operations Manager – Student Services
- Purpose of use: To simulate stakeholder interview for requirement elicitation
- Date used: 9/2/2026
- Conversation link: https://chat.deepseek.com/share/c0pfcjv1jjwoqd78ip

2. Task Description

DeepSeek was used to simulate an interview with an Operations Manager due to the absence of access to real stakeholders. The objective was to gather qualitative business goals, operational challenges, KPIs, and expectations for a future system.

3. Initial Prompt

Act as the Operations Manager of ABC University Student Services. Your responsibilities include overseeing enquiry management, staff performance, service quality, and operational efficiency. You are being interviewed by a Business Analyst about the current enquiry management system. Please describe:
1. Your key business goals.
2. Major operational challenges.
3. Current inefficiencies in the enquiry handling process.
4. Performance metrics you care about (KPIs).
5. Risks and compliance concerns.
6. Your expectations for a future improved system.
Be realistic, detailed, and business-focused. Do not provide technical solutions unless asked.

4. Initial AI Output (Excerpt)

Core Mission
To drive student retention and success by providing seamless, empathetic, and proactive support while optimizing university resources.

Current Pain Points
- Information Silos: Data is scattered across multiple systems, preventing a "single view" of the student.
- Manual Workflows: Triage and routing are done manually, leading to delays and "hand-off" failures.
- High Pressure: Extreme volume peaks and high staff turnover result in inconsistent service and burnout.
- Outdated Tools: Static FAQs and a lack of self-service options force staff to handle repetitive, simple queries.

Key Performance Indicators (KPIs)
- First Contact Resolution (FCR): The primary metric for efficiency.
- Resolution Time & CSAT: Measuring speed and student satisfaction.

Critical Risks
- Compliance: Potential FERPA (data privacy) breaches due to ad-hoc sharing.
- Inconsistency: Providing wrong policy information, leading to reputational damage.

The Solution: A Unified SRM Platform
The manager expects more than a ticketing system; they need a Student Relationship Management (SRM) platform that offers:
- 360-Degree Student Dashboard: Integrated data from all departments.
- Automation: Intelligent routing and automated triage.
- Knowledge Base: A central, dynamic repository for accurate policy info.
- Self-Service Portal: Empowering students to find answers and track cases independently.

5. Iterative Prompting Process

Follow-up Prompt 1:

Can you quantify current enquiry volumes and average resolution time?

Reason for prompt:

The initial response lacked measurable KPIs.

AI Response Summary:

The data reveals a critical mismatch: while 65% of our volume (Tier 1 & 2) should be resolvable in under 24 hours, our actual average resolution time for email (our highest volume channel) is over 52 hours. This gap represents a significant operational deficit, directly impacting student satisfaction and staff workload. The low email FCR rate points directly to the root causes: inefficient triage, lack of a unified knowledge base, and information silos forcing staff to "work around" systems rather than working effectively within them.

Follow-up Prompt 2:

If budget and time were limited, which three capabilities would you prioritize for phase one implementation, and why?

Reason for prompt:

This question helps me to define MVP and Must-have/Nice-to-have

AI Response Summary:

The goal is to deliver maximum impact with constrained resources by focusing on foundational efficiency rather than complex features.

1. Centralized Case Management & Intelligent Routing
Purpose: To eliminate the "black hole" of manual email triaging.
Impact: Cuts resolution time by 30–40% and ensures every student inquiry has a clear owner and audit trail.
Value: Provides the first-ever real-time data on staff workload and bottlenecks.

2. Integrated Knowledge Base
Purpose: To provide a "single source of truth" for all university policies.

Impact: Boosts First Contact Resolution (FCR) by 10–15% and significantly reduces training time for new hires.
Value: Ensures consistent, compliant advice across all service channels.

3. Unified Student Profile
Purpose: A simplified dashboard pulling data from Banner, Housing, and Financial Aid.
Impact: Staff instantly see a student's status and history, eliminating the need for students to repeat their stories.
Value: A high-impact "quick win" for student satisfaction without the complexity of a full CRM integration.

6. Human Evaluation & Refinement

The AI response was comprehensive but lacked specific technical constraints. I refined the requirements by:
- Prioritising unified student profile as Phase 1 deliverable
- Converting qualitative goals into measurable KPIs
- Removing overly generic AI suggestions not aligned with FERPA compliance

While the AI response was structured and comprehensive, several limitations were identified:
- Some performance metrics were hypothetical and not grounded in verified institutional data.
- The suggested impact percentages (e.g., 30–40% efficiency improvement) lacked empirical evidence.
- Technical integration assumptions (e.g., seamless Banner integration) may underestimate implementation complexity.
- Compliance considerations were high-level and required further validation against actual FERPA institutional policies.

Therefore, human judgment was applied to filter speculative claims and ensure feasibility and alignment with real-world university constraints.

7. Final Use of AI Output

Selected and critically evaluated AI-generated insights were used to:

- Develop functional requirements
- Construct epics and use case diagram
- Identify compliance risks for non-functional requirements
- Support literature discussion on AI-assisted triage systems

AI was used solely for stakeholder simulation and initial idea generation. All system models (epics, use case diagrams, activity diagrams, ERD) were independently designed, validated, and structured by the author based on extracted and critically evaluated requirements.

II. Copilot

1. AI Tool Information

- Tool name: Copilot
- Role simulated: Administrative Officer – Student Services
- Purpose of use: To simulate frontline operational perspective in enquiry handling
- Date used: 10/2/2026
- Conversation link: https://copilot.microsoft.com/shares/c2v1jzEcvqxgczJP3mCLY

2. Task Description

Copilot was used to simulate an Administrative Officer working at the Student Services frontline.The objective was to: Understand day-to-day operational workflows; identify system usability problems; capture pain points in enquiry handling; validate managerial-level insights from DeepSeek; identify functional requirements from staff perspective

This complements the strategic-level interview (Operations Manager) by adding operational detail and practical constraints.

3. Initial Prompt

Act as an Administrative Officer working at ABC University Student Services.
You handle student enquiries via walk-in, phone, and email daily.
You are being interviewed by a Business Analyst about the current enquiry management system.
Please describe:
- Your daily tasks and responsibilities
- How you currently handle student enquiries step-by-step
- What tools or systems you use
- Pain points and frustrations in your workflow
- Situations where students get upset or frustrated
- What features would make your job easier
Be realistic and operational. Focus on actual day-to-day work experience rather than strategy.

4. Initial AI Output (Excerpt)

1. Core Responsibilities & Workflow
- Multichannel Support: Managing the "Front-of-House" (walk-ins), phone hotline, and shared email inboxes.
- The 8-Step Lifecycle: A systematic approach from Intake → Verification → Resolution (Simple) or Escalation (Complex) → Follow-up → Closure.
- Liaison Role: Acting as the bridge between students and specialized departments (Finance, Housing, IT).

2. The "Manual Burden" (Tools & Systems)
- Fragmented Toolkit: Reliance on a basic CRM, shared Outlook inboxes, and manual Excel spreadsheets.

- Paper Trails: Critical processes (like Leave of Absence) are still handled via paper forms, causing delays and data entry errors.

3. Key Pain Points
- Information Blindness: Once an issue is escalated to another department, frontline staff lose visibility, making it impossible to give students accurate status updates.
- Data Redundancy: Staff frequently enter the same data into multiple systems (CRM + Excel + Paper).
- Peak Volume Stress: System crashes and high enquiry volumes during registration lead to "response time lag" and student burnout.

4. Root Causes of Student Anger
- Inconsistency: Receiving conflicting instructions from different departments.
- Process Friction: Portal crashes during high-stakes periods (registration/results).
- Expectation Gaps: Students often struggle with rigid policies regarding fees and deadlines.

5. The "Ease-of-Life" Wishlist
- Unified Platform: Merging student records and departmental workflows into one screen.
- Student Self-Service: A portal where students can track their own case status to reduce "status check" calls.
- Smart Automation: Auto-routing enquiries to the correct department and using an analytics dashboard to predict peak times.

5. Iterative Prompting Process

Follow-up Prompt 1:

In a typical peak period (e.g., enrolment week), how many enquiries do you handle per day?
What percentage are simple vs complex?
How often do you need to escalate cases?

Reason for prompt:

To quantify workload and support workload analysis + system capacity planning.

AI Response Summary:

During peak periods like enrolment week, an Administrative Officer manages a heavy volume of 120–150 enquiries daily across walk-ins, calls, and emails. While 60–70% are simple queries resolved instantly, the remaining 30–40% are complex issues requiring deep investigation. The main operational bottleneck occurs with the 20–25% of cases that must be escalated to other departments; due to a lack of integrated tracking, staff lose visibility once a case leaves their hands, leading to student frustration and repetitive follow-up calls that further inflate the workload.

Follow-up Prompt 2:

Describe a recent situation where a student complaint escalated due to system inefficiency. What caused it?

Reason for prompt:

To identify failure points and support risk analysis and non-functional requirements.

AI Response Summary:

This case study illustrates the direct consequences of poor system integration: when a student encountered a "course full" error due to a sync lag between the portal and the registry database, staff were forced into a slow, manual email workaround. Because the student had zero visibility into the progress of their inquiry, they repeatedly returned to the counter in a panic, eventually escalating the complaint to the Head of Student Services. This scenario underscores how technical silos transform simple glitches into significant service failures, causing unnecessary stress for students and wasting hours of staff time on manual follow-ups and complaint management.

6. Human Evaluation & Refinement

While the AI-generated responses provided detailed operational insights, several limitations were identified:
- Workload numbers were estimated and not based on real institutional data.
- The AI assumed uniform system usage across departments.
- Emotional impact and staff burnout descriptions were generalized.
- Escalation procedures were described without reference to formal policy structures.

Therefore, human judgment was applied to:
- Align operational insights with managerial KPIs gathered earlier.
- Remove exaggerated claims.
- Convert narrative frustrations into structured functional requirements.
- Cross-check compliance implications (FERPA, data access controls).

Additionally, AI-generated operational scenarios were treated as hypothetical and were not assumed to reflect real institutional practices without validation.

7. Final Use of AI Output

AI-generated operational insights were selectively used to:

- Refine functional requirements (e.g., auto-assignment, escalation workflow)
- Develop activity diagrams for enquiry handling
- Identify usability and system performance non-functional requirements
- Validate bottlenecks identified at managerial level
- Support risk and compliance modelling

AI was used for role simulation only. These insights directly informed the Activity Diagram for Enquiry Lifecycle and the ERD design of Case, Escalation, and Interaction entities. All system models, requirements structuring, and prioritisation decisions were independently analysed and documented by the author.

III. Gemini

1. AI Tool Information

- Tool name: Gemini
- Role simulated: Current Undergraduate Student – ABC University
- Purpose of use: To simulate end-user perspective in enquiry and student support experience
- Date used: 10/2/2026
- Conversation link: https://gemini.google.com/share/d5a210a90543

2. Task Description

Gemini was used to simulate a current undergraduate student interacting with Student Services. The objective was to: Understand real student experiences when seeking support; identify emotional pain points and service gaps; capture usability issues in portals and enquiry processes; validate operational inefficiencies identified earlier and extract user-centered functional and non-functional requirements

This complements:
Strategic perspective (Operations Manager – DeepSeek)
Operational perspective (Administrative Officer – Copilot)
By adding the end-user experience dimension.

3. Initial Prompt

Act as a second-year undergraduate student at ABC University.
You have interacted with Student Services multiple times regarding enrolment, financial aid, academic advice, and administrative issues.
You are being interviewed by a Business Analyst about your experience with the current enquiry and support system.
Please describe:
- Why you usually contact Student Services
- How you submit enquiries (walk-in, phone, email, portal)
- Your experience step-by-step when seeking help
- Frustrations or delays you encountered
- Situations where you felt stressed or unsupported
- What features would improve your experience
Be honest, realistic, and describe emotional as well as practical aspects. Focus on user experience rather than technical solutions.

4. Initial AI Output (Excerpt)

Here is the outline of the student's perspective:

1. Motivation: High-Stakes Issues
- Critical Needs: Primarily contacts the university for "make or break" issues like financial aid, scholarship clearance, and enrollment glitches.
- Academic Planning: Needs guidance on complex changes, such as swapping minors without delaying graduation.

2. The Student Journey: "The Maze"
- Multi-Channel Desperation: Starts with the Portal, moves to Email for a human touch, and resort to Walk-ins only for emergencies.
- The "Void": After receiving an instant auto-reply, students wait 3–5 business days with no status updates.
- Departmental Ping-Pong: Frequently told to contact a different department (e.g., from Enrolments to Finance), forcing them to restart the process from scratch.

3. Key Frustrations
- Fragmented Systems: Departments (Finance, Faculties, Student Life) do not share information, leading to conflicting advice.
- Authority Gaps: Staff can see the problem in the system but often lack the permission/tools to click "approve" or fix it immediately.
- The Emotional Toll: Navigating the "maze" while stressed about exams makes students feel like a nameless ID number rather than a person.

4. The Student Wishlist
- Single Source of Truth: A unified dashboard to track the progress of an enquiry across all departments.
- Transparency: Real-time "Estimated Wait Times" to manage expectations and reduce anxiety.
- Human Connection: Access to short Zoom calls for complex issues to avoid endless, confusing email chains.

5. Iterative Prompting Process

Follow-up Prompt 1:

During peak enrolment week, how long did you typically wait for responses? How did that delay affect your academic decisions?

Reason for prompt:

To quantify service delays and identify impact on student retention and decision-making.

AI Response Summary:

Peak enrollment week creates an "academic bottleneck" where typical 3-day response times swell to 10 days, leaving students in a stressful state of uncertainty. The total "silence" from the university during this period triggers a domino effect on their studies: by the time approvals or prerequisite waivers finally arrive, preferred class slots are often full, and critical deadlines for changing majors have already passed. This "planning paralysis" prevents students from committing to timetables or purchasing materials, ultimately forcing them to be reactive and settle for "whatever is left over" rather than proactively choosing their own educational path.

Follow-up Prompt 2:

Have you ever submitted multiple enquiries about the same issue? Why?

Reason for prompt:

To identify duplication, lack of transparency, and system trust issues.

AI Response Summary:

Students often contribute to system congestion by submitting multiple tickets for the same issue, a behavior driven by "black hole" anxiety and a total lack of transparency regarding their initial request's status. This desperation leads to a "priority escalation" strategy, where students try different channels, categories, or even a second ticket in hopes of finding a more helpful staff member or a faster response. The disconnect between digital and in-person support further complicates this, as walk-in visits often create "secondary flags," leaving students feeling exhausted and forced to "poke" the system repeatedly just to ensure they aren't ignored in a digital room where only the loudest get noticed.

6. Human Evaluation & Refinement

While the AI-generated student responses provided realistic user experience narratives, several limitations were identified:
- Emotional intensity may be exaggerated
- Response times were estimated, not verified
- AI assumed homogeneous student experience
- Policy knowledge was simplified

Therefore, human judgment was applied to:
- Cross-check response times with managerial KPIs
- Remove over-dramatised emotional language
- Convert emotional pain points into measurable requirements
- Identify actual system features required rather than generic complaints

Additionally, AI-generated student feedback was treated as simulated user research data, not empirical evidence.

7. Final Use of AI Output

Selected insights from Gemini were used to:

- Develop user-centered functional requirements
- Define usability non-functional requirements (e.g., response time < 24 hours)
- Construct student actor in use case diagram
- Design activity diagram from user initiation perspective
- Identify system transparency requirements (case tracking, notification system)
- Strengthen justification for self-service portal features

These insights directly informed the design of the Student Portal use cases, notification workflow in the activity diagram, and the CaseStatus entity in the ERD. AI was used solely for stakeholder simulation. All requirement validation, modelling, and prioritisation decisions were independently conducted by the author.

IV. Functional Requirement Table

ID	Functional Requirement	Source	Priority
FR-01	The system shall allow students to submit enquiries via an online portal.	Student	High
FR-02	The system shall generate a unique case ID for each enquiry.	Admin Officer	High
FR-03	The system shall send automatic confirmation notifications upon submission.	Student	High
FR-04	The system shall allow staff to manually create cases for walk-in or phone enquiries.	Admin Officer	High
FR-05	The system shall automatically assign cases based on enquiry category.	Manager	High
FR-06	The system shall allow staff to reassign cases to another department.	Admin Officer	High
FR-07	The system shall allow escalation of cases with tracking history.	Admin Officer	High
FR-08	The system shall maintain a complete audit trail of case interactions.	Manager	High
FR-09	The system shall allow students to track case status in real time.	Student	High
FR-10	The system shall display estimated response time for each case.	Student	Medium
FR-11	The system shall notify students when case status changes.	Student	High
FR-12	The system shall display a unified student profile to authorized staff.	Manager	High
FR-13	The system shall display previous case history for each student.	Admin Officer	High
FR-14	The system shall provide a centralized knowledge base for staff.	Admin Officer	Medium
FR-15	The system shall allow knowledge articles to be updated by authorized users.	Manager	Medium
FR-16	The system shall allow students to search FAQs via the portal.	Student	Medium
FR-17	The system shall generate performance reports (FCR, resolution time).	Manager	High
FR-18	The system shall provide dashboard analytics for enquiry volume by category.	Manager	Medium
FR-19	The system shall allow secure messaging between students and staff within each case.	Student	High
FR-20	The system shall allow internal notes visible only to staff.	Admin Officer	High

V. Non-Functional Requirement Table

ID	Category	Requirement
NFR-01	Performance	The system shall respond to user actions within 2 seconds under normal load conditions.
NFR-02	Performance	The system shall support at least 500 concurrent users during peak enrolment periods.
NFR-03	Performance	Email or portal enquiries shall be logged into the system within 5 seconds of submission.
NFR-04	Security	The system shall enforce role-based access control (RBAC) for all users.
NFR-05	Compliance	The system shall comply with FERPA regulations regarding student data privacy.
NFR-06	Security	All communication between users and the system shall be encrypted (HTTPS).
NFR-07	Security	The system shall maintain audit logs of all case updates and data access.
NFR-08	Availability	The system shall maintain at least 99% uptime annually.
NFR-09	Reliability	The system shall automatically back up data daily.
NFR-10	Reliability	The system shall recover from system failure within 30 minutes.
NFR-11	Usability	The student portal shall be accessible via desktop and mobile devices.
NFR-12	Usability	Users shall be able to submit an enquiry in no more than 5 steps.
NFR-13	Usability	The system shall provide clear status labels (e.g., Open, In Progress, Escalated, Closed).
NFR-14	Scalability	The system shall support increasing enquiry volume without performance degradation.